

Lecture-#1

Natural Language Processing

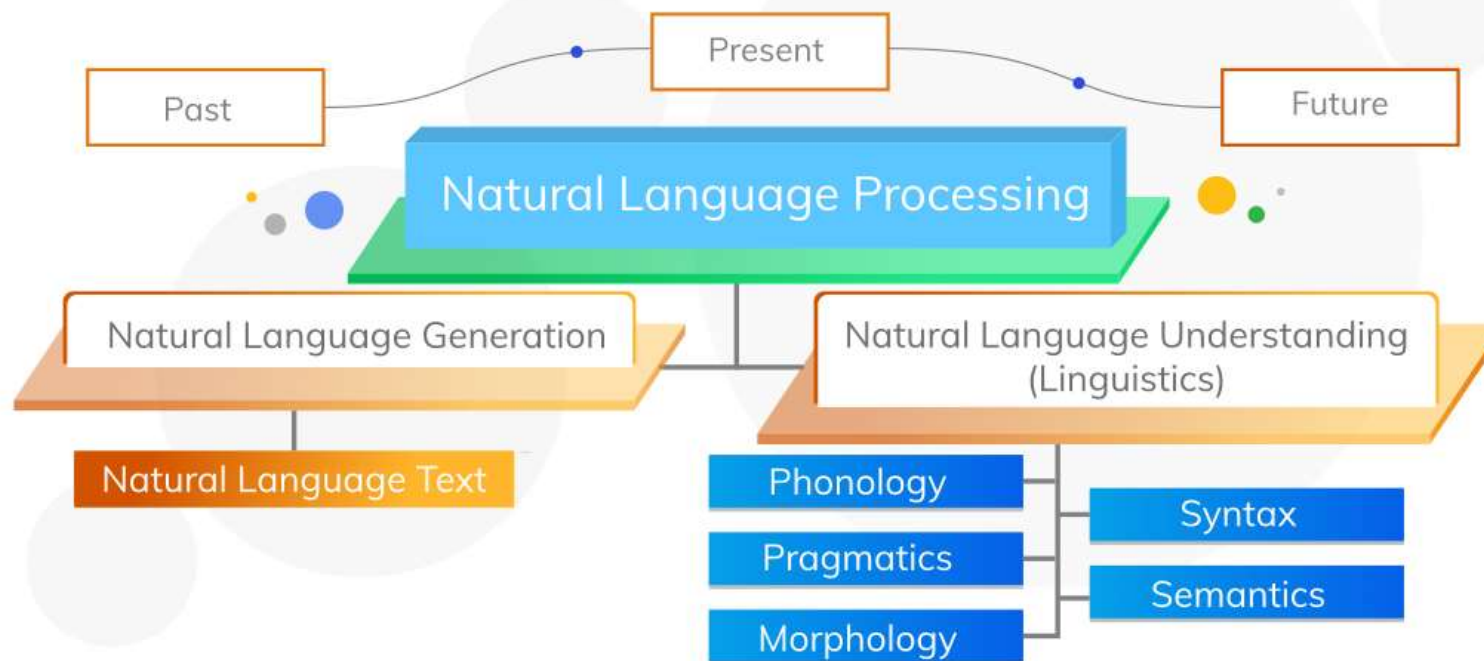
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Introduction

Origins of Natural Language Processing

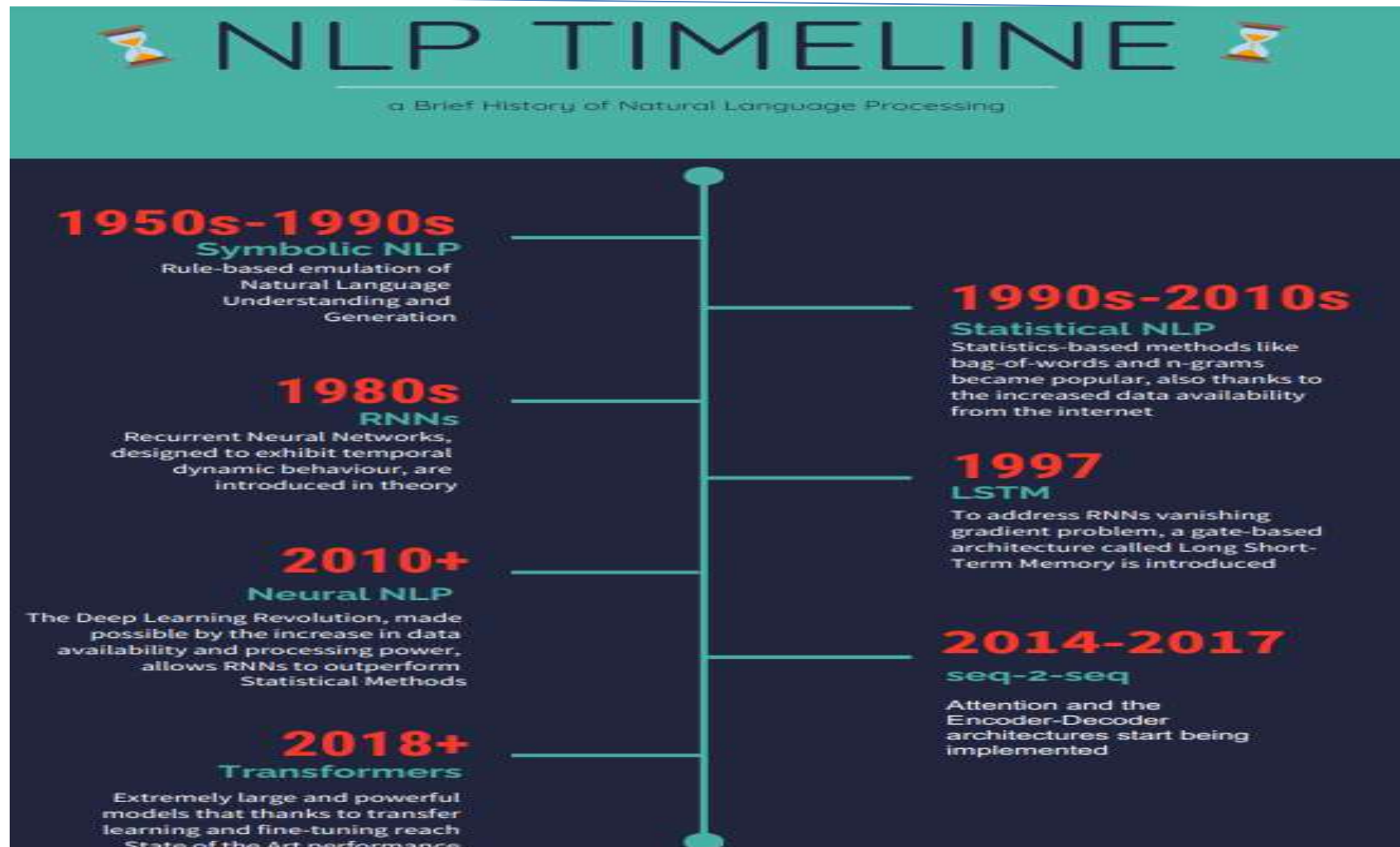


Evolution of NLP



Origins of Natural Language Processing

Alan Turing is the father of Natural language processing. In his **1950** paper **Computing Machinery and Intelligence**, he described a test for an intelligent machine that could understand and respond to natural human conversation.



NLP Terminology

NLP Terminology

Phonology – It is study of organizing sound systematically.

Morphology: The study of the formation and internal structure of words.

Morpheme – It is primitive unit of meaning in a language.

Syntax: The study of the formation and internal structure of sentences.

Semantics: The study of the meaning of sentences.

Pragmatics – It deals with using and understanding sentences in different situations and how the interpretation of the sentence is affected.

Discourse – It deals with how the immediately preceding sentence can affect the interpretation of the next sentence.

World Knowledge – It includes the general knowledge about the world.

Phases of NLP

1. Lexical Analysis and Morphological

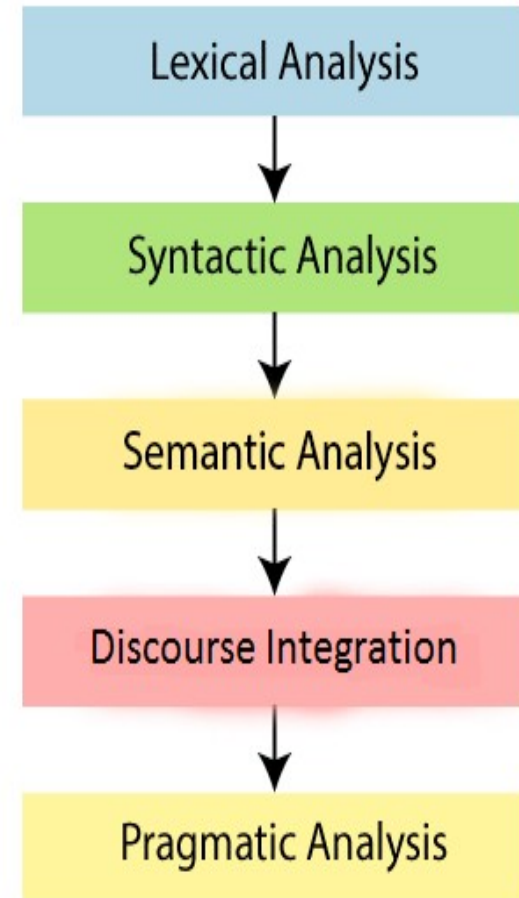
- The first phase of NLP is the Lexical Analysis. This phase scans the source code as a stream of characters and converts it into meaningful lexemes.
- It divides the whole text into paragraphs, sentences, and words.

2. Syntactic Analysis (Parsing)

- Syntactic Analysis is used to check grammar, word arrangements, and shows the relationship among the words.

Example: Agra goes to the Poonam

- In the real world, Agra goes to the Poonam, does not make any sense, so this sentence is rejected by the Syntactic analyzer.



Phases of NLP

3. Semantic Analysis

• Semantic analysis is concerned with the meaning representation. It mainly focuses on the literal meaning of words, phrases, and sentences.

4. Discourse Integration

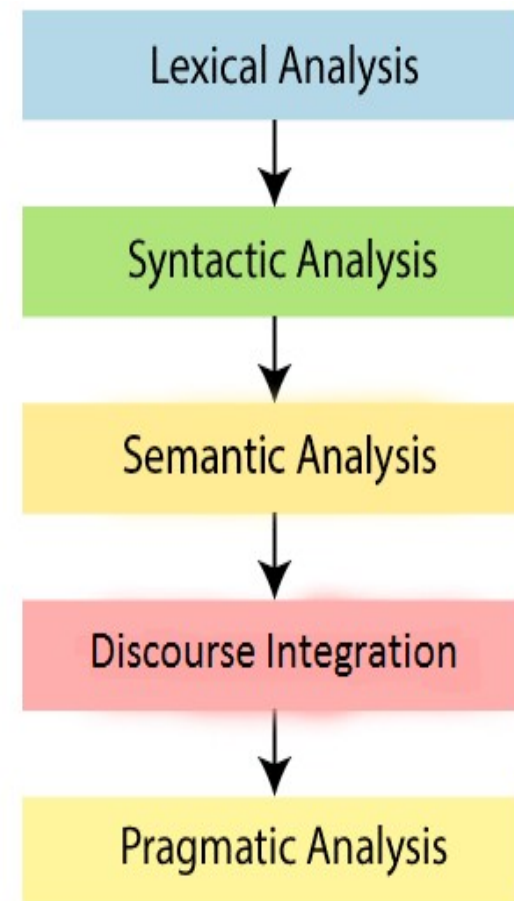
• Discourse Integration depends upon the sentences that precedes it and also invokes the meaning of the sentences that follow it.

• **For example**, the word "that" in the sentence "He wanted that" depends upon the prior discourse context.

5. Pragmatic Analysis

• Pragmatic is the fifth and last phase of NLP. It helps you to discover the intended effect by applying a set of rules that characterize cooperative dialogues.

For Example: "Open the door" is interpreted as a request instead of an order.



The Challenges of NLP

❖ NLP is a powerful tool with huge benefits, but there are still a number of Natural Language Processing limitations and problems:

Contextual words and phrases and homonyms

Synonyms

Ambiguity

Errors in text or speech

Colloquialisms and slang

The Challenges of NLP

❖ Contextual words and phrases and homonyms

1. The same words and phrases can have different meanings according to the context of a sentence and many words – especially in English – have the exact same pronunciation but totally different meanings.

2. Homonyms – two or more words that are pronounced the same but have different definitions – can be problematic for question answering and speech-to-text applications because they aren't written in text form. Usage of *their* and *there*, for example, is even a common problem for humans.

For example:

*I **ran** to the store because we **ran** out of milk.*

*Can I **run** something past you real quick?*

*The house is looking really **run** down.*

These are easy for humans to understand because we read the context of the sentence and we understand all of the different definitions. And, while NLP language models may have learned all of the definitions, differentiating between them in context can present problems.

The Challenges of NLP



Learning from Experience, Sharing with Passion

❖ Synonyms

Synonyms can lead to issues similar to contextual understanding because we use many different words to express the same idea.

Furthermore, some of these words may convey exactly the same meaning, while some may be levels of complexity (small, little, tiny, minute) and different people use synonyms to denote slightly different meanings within their personal vocabulary.

So, for building NLP systems, it's important to include all of a word's possible meanings and all possible synonyms.

Text analysis models may still occasionally make mistakes, but the more relevant training data they receive, the better they will be able to understand synonyms.

The Challenges of NLP

❖ Ambiguity

Ambiguity in NLP refers to sentences and phrases that potentially have two or more possible interpretations.

Lexical ambiguity: a word that could be used as a verb, noun, or adjective. **Example** Light, Fast

Semantic ambiguity: the interpretation of a sentence in context. For example: *I saw the boy on the beach with my binoculars.*

This could mean that I saw a boy through my binoculars or the boy had my binoculars with him

Syntactic ambiguity: In the sentence above, this is what creates the confusion of meaning. The phrase *with my binoculars* could modify the verb, “saw,” or the noun, “boy.”

Even for humans this sentence alone is difficult to interpret without the context of surrounding text.

POS (part of speech) tagging is one NLP solution that can help solve the problem, somewhat.

The Challenges of NLP



Learning from Experience, Sharing with Passion

❖ Colloquialisms and slang

Informal phrases, expressions, idioms, and culture-specific lingo present a number of problems for NLP – especially for models intended for broad use.

Because as formal language, colloquialisms may have no “dictionary definition” at all, and these expressions may even have different meanings in different geographic areas.

Furthermore, cultural slang is constantly morphing and expanding, so new words pop up every day.

This is where training and regularly updating custom models can be helpful, although it oftentimes requires quite a lot of data.

Language and Grammar

Difference between Natural language and Computer Language

Natural Language	Computer Language
Natural language has a very large vocabulary.	Computer language has a very limited vocabulary.
Natural language is easily understood by humans.	Computer language is easily understood by the machines.
Natural language is ambiguous in nature.	Computer language is unambiguous.

❖ NLP is usually used for chatbots, virtual assistants, and modern spam detection.

❖ But NLP isn't perfect, although there are over 7000 languages spoken around the globe, most NLP processes only use seven languages: English, Chinese, Urdu, Farsi, Arabic, French, and Spanish.

Language and Grammar

❖ Grammar in NLP is a set of rules for constructing sentences in a language used to understand and analyze the structure of sentences in text data.

Mathematically, a grammar G can be written as a 4-tuple (N, T, S, P) where,

N or V_N = set of non-terminal symbols, or variables.

T or Σ = set of terminal symbols.

S = Start symbol where $S \in N$

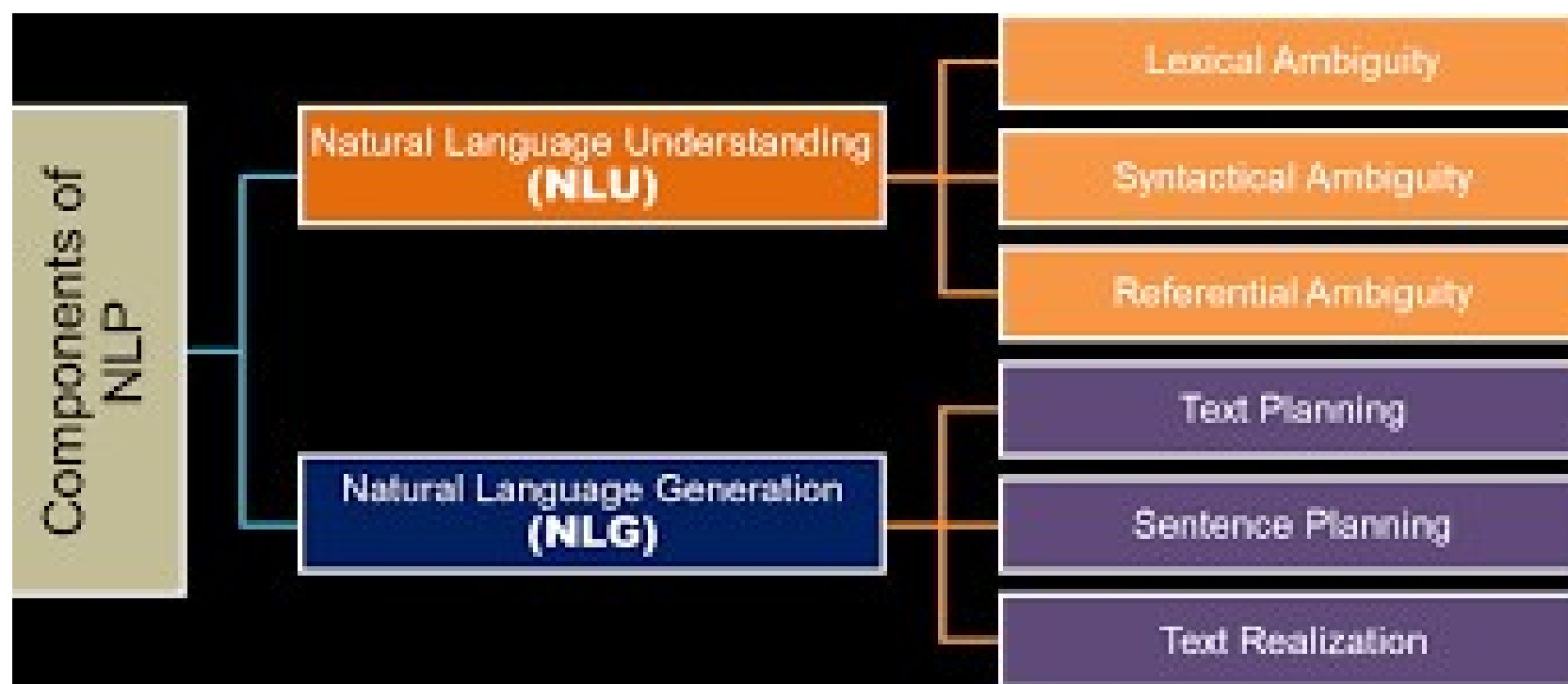
P = Production rules for Terminals as well as Non-terminals.

It has the form $\alpha \rightarrow \beta$, where α and β are strings on $V_N \cup \Sigma$ and at least one symbol of α belongs to V_N



Components of NLP

There are the following two components of NLP –



Components of NLP

Natural language understanding (NLU)

- ❖ NLU enables machines to understand and interpret human language by extracting metadata from content. It performs the following tasks:
- ❖ Helps analyze different aspects of language.
- ❖ Helps map the input in natural language into valid representations.
- ❖ NLU is more difficult than NLG tasks owing to referential, lexical, and syntactic ambiguity.
- ❖ **Lexical ambiguity:** This means that one word holds several meanings. For example, "The man is looking for the match." The sentence is ambiguous as 'match' could mean different things such as a partner or a competition.
- ❖ **Syntactic ambiguity:** This refers to a sequence of words with more than one meaning. For example, "The fish is ready to eat." The ambiguity here is whether the fish is ready to eat its food or whether the fish is ready for someone else to eat. This ambiguity can be resolved with the help of the part-of-speech tagging technique.
- ❖ **Referential ambiguity:** This involves a word or a phrase that could refer to two or more properties. For example, Tom met Jerry and John. They went to the movies. Here, the pronoun 'they' causes ambiguity as it isn't clear who it refers to.

Components of NLP

Natural language generation (NLG)

❖ NLG is a method of creating meaningful phrases and sentences (natural language) from data. It comprises three stages: text planning, sentence planning, and text realization.

❖ **Text planning:** Retrieving applicable content.

❖ **Sentence planning:** Forming meaningful phrases and setting the sentence tone.

❖ **Text realization:** Mapping sentence plans to sentence structures.

❖ [Chatbots](#), machine translation tools, analytics platforms, voice assistants, sentiment analysis platforms, and AI-powered transcription tools are some applications of NLG.

Components of NLP

NLU	NLG
NLU is the process of reading and interpreting language.	NLG is the process of writing or generating language.
It produces non-linguistic outputs from natural language inputs.	It produces constructing natural language outputs from non-linguistic inputs.

